

Topology with long distance

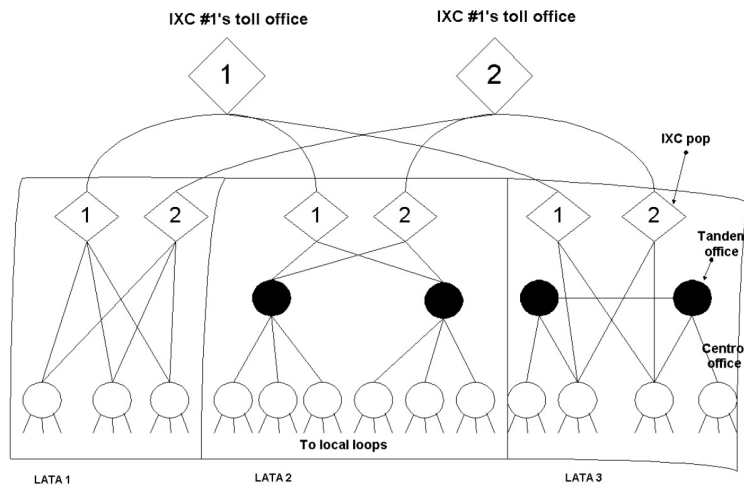


Fig. 2-16. The relationship of LATAs, LECs, and IXC's. All the circles are LEC switching offices. Each diamond belongs to the IXC whose number is in it

Multiple long-distance carriers

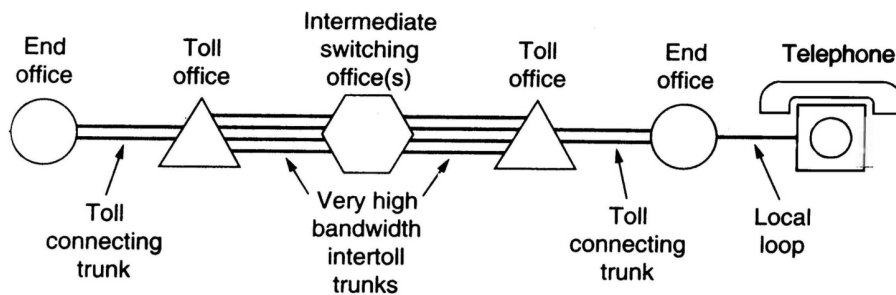
LATA = Local Access and Transport Area

LEC = local exchange carrier

IXC = Interexchange (long distance) carrier

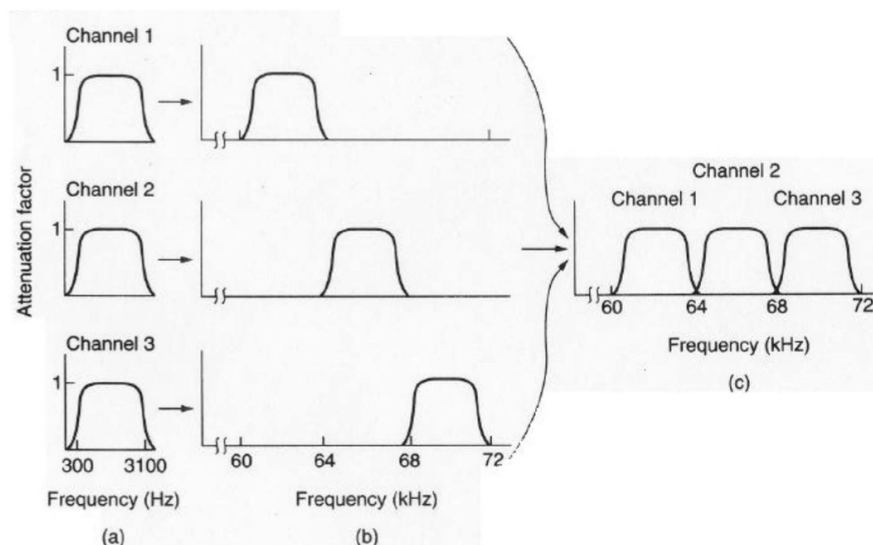
LEC $\sim$ LATA $\rightarrow$ IXC $\rightarrow$ End office $\rightarrow$ Telephone

POP point of presence



Tanenbaum fig. 2-15

FDM



TDM

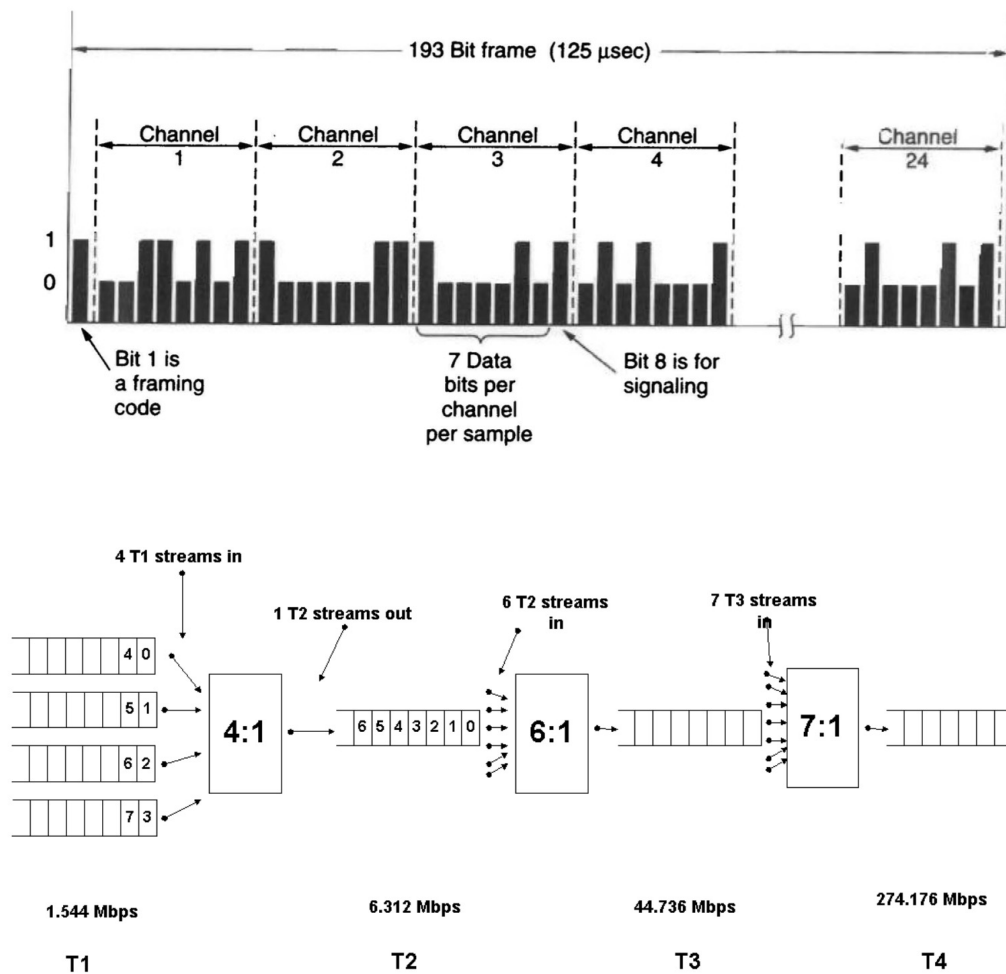


Fig. 2-28 Multiplexing T1 streams onto higher carriers

Circuit switching

Once a call has been set up, a dedicated path between both ends exists and will continue to exist until the call is finished.

propagation delay $\propto d^2$, can't arrive out of order

congestion time: setup time

wasted

packet switching

There is no need to set up a dedicated path in advance.

Router store and forward. Can arrive out of order.

congestion time: packet sent time

route deadend

connect time is not an issue, volume of traffic is.

Simplicity and efficiency